6.2 The Natural Base e

Algebra II 6.2 · Student sheet

Name _____ Period ____ Date _____

I can…

- define and use the natural base e.
- graph natural base functions and tell growth from decay.
- solve real-life problems with continuously compounded interest.

Vocabulary · fill in the missing word

Natural base e — an _____ number, $e \approx 2.71828$, the value $(1 + \frac{1}{n})^n$ approaches as n gets very large

Natural base exponential function — $y = ae^{rx}$: _____ when $a > 0$ and $r > 0$, decay when $a > 0$ and $r < 0$

Continuously compounded interest — $A =$ _____

Part 1 · Compound more often

One dollar earns 100% interest for one year, compounded n times. Find $(1 + 1/n)^n$ to four decimal places.

n	1	2	4	12	365	8,760
$(1 + 1/n)^n$						

Now type e^1 on your calculator: _____ What do you notice?

Part 2 · Notes and examples (fill in each note, then work the example)

The natural base e: $e \approx$ _____. It is the value $(1 + 1/n)^n$ approaches as n _____.

Example 1

Simplify each expression.

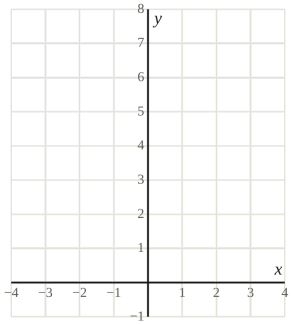
a. $e^3 \cdot e^5$ b. $\frac{16e^5}{4e^4}$ c. $(3e^{-4x})^2$

Growth or decay: $y = ae^{rx}$ $(a > 0)$ shows growth when _____ and decay when _____.

Example 2

Tell whether each function represents exponential growth or decay. Then graph them.

a. $y = 3e^{2x}$ b. $y = e^{(-0.5x)}$



Rewriting: $e^{rt} = (e^r)^t$, so the growth factor is _____.

Example 3

Rewrite $y = 2e^{(0.06t)}$ in the form $y = a(1 + r)^t$. Find the percent rate of change.

Continuously compounded interest: A = _____

Example 4

You deposit $4,000 in an account that pays 5% annual interest compounded continuously. Find the balance after 6 years.

Example 5

Find the balance for the same $4,000 at 5% for 6 years compounded quarterly. Which account earns more?

Part 3 · Partner practice

Partner 1

Simplify $(2e^{3x})^3$.

Partner 2

You deposit $2,500 at 4% annual interest compounded continuously. Find the balance after 10 years.